

- **Branching**

- Take three numbers and print the largest among them.
- Given a number, determine if it is a 3-digit palindrome

- **Loops**

- Print the given series using a loop: 1, -3, 5, -7, 9, -11, 13, -15
- Find factorial of N
- Find out maximum and minimum from N inputs
- Print the Fibonacci series up to N terms.
- Reverse an input integer (not allowed to use Strings). Input=> 12345, Output => 5, 4, 3, 2, 1
- Convert binary to integer (not allowed to use Strings). Input=> 10111010, Output => 186
- Count even digits (not allowed to use Strings). Input=> 12345, Output => 2 even digits
- Largest digit of input (not allowed to use Strings). Input=> 5375264, Output => 7 is largest

- **Nested Loop**

- Print all prime numbers from a specific range where the range will be taken as input.
- Print all perfect numbers from a specific range where the range will be taken as input. (A number is perfect if sum of its proper divisors equals the number, e.g., 6 = 1+2+3)
- Find out the sum of digits of N integer inputs and print the input with the largest sum. (If N = 5, calculate the sum of digits for all 5 inputs and print the input where the sum of digits is largest).

- **String [Allowed methods: length(), charAt(), codePointAt(), equals(), next(), nextLine()]**

- Only print the lowercase characters
- Convert all characters to lowercase
- Convert the lowercase character to uppercase and vice versa
- Remove consecutive duplicates
- Write a Java program that takes a person's first name and email's password as user inputs and checks whether that password is valid or not. A password will be considered invalid if the person's name is present anywhere in that password. *You can assume that both the name & password will not contain any uppercase characters and the length of the password will always be greater than the length of name.* **[Note: You are not allowed to use the built-in contains() or substring() method to solve this problem.]**

| Sample Input                  | Output  |
|-------------------------------|---------|
| dipu<br>iamdipu123#           | Invalid |
| shajib<br>happy new year 2025 | Valid   |
| dipu<br>iamdip123#            | Valid   |

- You are a class teacher at a kindergarten school. As a task, you asked your students to write a paragraph. Unfortunately, you will notice that most of the students did not use capital letters correctly. Your task is to write a java program that takes a string as input and prints a new copy of the string that has been correctly capitalized. Your code should:
  - Capitalize the first letter in the string
  - Capitalize the first letter after a full-stop, exclamation mark, or question mark
  - Capitalize the word “i” if it is in lowercase.

| Input                                                                                                                                             | Output                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| my favourite animal is a dog. a dog has sharp teeth so that it can eat flesh very easily. do you know my pet dog's name? i love my pet very much. | My favourite animal is a dog. A dog has sharp teeth so that it can eat flesh very easily. Do you know my pet dog's name? I love my pet very much. |

- **Array**

- Find out the minimum value and its index from an array
- Create an array with size N which only takes negative values and calculate the sum and average of the array
- Index-wise sum of two arrays where arrays can be of different length
- Remove duplicates
- Remove consecutive duplicates
- Increase array size by N
- Trace Bubble Sort and Selection Sort codes
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- **Methods**

- All loop, string and array tasks